

Project distributions

Knowledge Representation

#	Task	Person
1	Define domain, scope, and initial competency questions (at least 20 covering taxonomy, properties, pragmatics, dataset)	All (together)
2	Enumerate important terms (road sign, shape, colour, symbol, pedestrian, lorry, etc.)	All (together)
3	Create formal taxonomy of road signs from Vienna Convention Part I, Annex 1 (IS-A hierarchy in OWL2, RDF/XML)	Gússem
4	Add disjointness axioms between sibling classes in the taxonomy	Gússem
5	Download and explore MTSD.ttl file; understand its class labels and structure	María



WEEK 2 (May 13–19)

#	Task	Person
6	Define object properties (hasShape, hasGroundColour, hasBorderColour, hasSymbol, etc.) with domain and range	Gússem
7	Define data properties (speedLimitValue, maxHeight, maxWeight, textInfo, plus MTSD metadata like width, height, isPano)	Gússem
8	Align Vienna Convention classes with MTSD labels using owl:equivalentClass (exact match)	María
9	Align remaining MTSD labels using rdfs:subClassOf (partial match) and add missing subclasses	María
10	Install Owlready2 with Pellet support; write basic script to load the ontology	Sara



WEEK 3 (May 20–26)

#	Task	Person
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11	Add property characteristics (functional, transitive, symmetric, etc.)	Gússem
12	Add class definitions using Manchester Syntax (necessary & sufficient conditions)	Gússem
13	Run Hermit or Pellet reasoner; check consistency and review inferred axioms	Gússem
14	Formulate DL queries for competency questions (paste into Protégé's DL Query tab)	Gússem
15	Formulate SPARQL queries for dataset-related competency questions	María
16	Write Python scripts for competency questions requiring aggregation or inference (e.g., percentages)	Sara

 WEEK 4 (May 27–31)

#	Task	Person
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17	Extract EL fragment of the ontology (focus on a subhierarchy like ProhibitoryOrRestrictiveSign)	Sara
18	Define probabilistic GCIs using B&L (Bayesian Description Logic) with contexts (e.g., city, rural, highway)	Sara
19	Adjust necessary conditions to different contexts (e.g., different shape standards for different countries)	Sara
20	Merge all components into final integrated ontology file	María
21	Run final reasoner check; verify all competency questions are answerable	All (together)
22	Compile documentation: for each task, record LLM used, prompt(s), revised output file	All (each documents their own tasks)

 Final Submission (Due 31 May)

#	Task	Person
23	Submit ontology file (RDF/XML or Turtle)	Gússem

24	Submit list of ≥ 20 competency questions	All (together)
25	Submit DL queries	Gússem
26	Submit SPARQL queries	María
27	Submit Python scripts	Sara
28	Submit EL fragment and probabilistic GCIs	Sara
29	Submit alignment documentation	María
30	Submit prompt records for all tasks	All (each submits their own)

Fundamentals of Natural Language

Week 1 (March 18 – April 7) – Analysis & Setup

#	Task	Person
1	Understand ICD coding task and the two paradigms (retrieval vs classification)	All
2	Obtain and explore the dataset: literals (diagnosis descriptions) and ground truth ICD codes	María
3	Extract ICD code descriptions (official titles) from ICD-10 taxonomy (e.g., from provided files like asho-uab-codification.pdf)	Sara

4	Analyze data characteristics: class imbalance, unique codes, literal length, abbreviations, misspellings	María
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 Week 2 (April 8 – May 10) – Implementation of Both Methods

#	Task	Person
7	Preprocessing pipeline: clean literals (lowercase, remove punctuation, handle abbreviations)	María
8	Method 1 (Semantic Similarity) – Word n-grams: vectorize literals and code descriptions using TF-IDF (word n-grams, e.g., 1-3)	Sara
9	Method 1 – Character n-grams: vectorize using character n-grams (e.g., n=3-5) to handle misspellings and morphology	Sara
10	Method 1 – Cosine similarity: compute similarity between each input literal and all code descriptions; return top-k matches	Sara

11	Method 1 – Evaluation: measure accuracy and macro-F1 for retrieval; experiment with different n-gram ranges	Sara
12	Method 2 (Supervised Classification) – Feature extraction: convert literals to TF-IDF vectors (train/test split)	Gússem
13	Method 2 – Model training: train linear SVM (or logistic regression) on literal-to-code pairs	Gússem
14	Method 2 – Hyperparameter tuning: cross-validation for C parameter, n-gram range	Gússem
15	Method 2 – Evaluation: accuracy and macro-F1 on test set	Gússem

 Week 3 (May 11 – May 31) – Analysis, Comparison & Finalization

#	Task	Person
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17	Compare both methods: accuracy, macro-F1, training time, inference time	María
18	Error analysis – Method 1: which codes are consistently mismatched? (e.g., rare codes, ambiguous literals)	María
19	Error analysis – Method 2: confusion matrix; which code pairs are frequently confused?	María
20	Qualitative analysis: show examples where one method succeeds and the other fails	María
21	Ablation experiment: compare word n-grams vs character n-grams for Method 1	Sara
22	Ablation experiment: compare SVM vs logistic regression for Method 2	Gússem

23	Write final report: introduction, methods, experiments, results, error analysis, conclusions	All (divide sections)
24	Prepare final presentation (10 minutes)	All